

Reranking, Context Selection, and Evidence Authority

Why Relevant Evidence Can Be Retrieved and Still Disappear

AI Memory and Reality Drift - Note #5 v1

A. Jacobs - Semantic Fidelity Project

The Basic Pattern

Retrieval usually produces more candidate passages than a language model can use. A reranker scores those candidates more carefully, and a context-selection step chooses the small set that enters the prompt.

This improves precision, but it creates an evidence bottleneck. A relevant source can be found and still have no effect because it ranks below the cutoff, duplicates a more popular passage, or consumes too many tokens. Storage and retrieval do not make information operational. Selection does.

Reranking is therefore more than technical cleanup. It allocates evidentiary authority.

When Ranking Determines What Can Matter

Cross-encoders, learned rerankers, fusion methods, metadata filters, and diversity rules all provide different representations of relevance. Their scores are translated into context position and inclusion.

Once the selected passages enter the prompt, the generator cannot distinguish between evidence that did not exist and evidence that was silently excluded. The final answer reflects the narrow context as though it were the available knowledge environment.

How Evidence Authority Emerges

Stage 1 - Candidate retrieval. A broad set of potentially relevant passages is found.

Stage 2 - Scoring. A reranker evaluates candidates against the query.

Stage 3 - Truncation. Token budgets and top-k limits remove lower-ranked material.

Stage 4 - Context assembly. Selected evidence is ordered and placed into the prompt.

Stage 5 - Authority. Inclusion becomes the practical boundary between evidence that can influence the answer and evidence that cannot.

Representational Authority and Reality Drift

Representational Authority is the power of a representation to organize later judgment and action. A reranking score gains this authority because it controls access to the model's working context.

Reality Drift emerges when the selected context remains coherent but ceases to reflect the important diversity, contradiction, or hierarchy of the source environment. The application remains grounded in retrieved text, yet the grounding is constrained by an invisible ranking surface.

The Representation Stack

Corpus becomes candidate set. Candidate set becomes relevance scores. Scores become a cutoff. The cutoff becomes context. Context becomes answer.

At every transition, the evidence environment narrows. A robust architecture should preserve rejected candidates, ranking explanations, source diversity, and a route for contradictory or authoritative evidence to bypass pure relevance ranking.

Examples Across AI Systems

Compliance: A general explanatory document may outrank the controlling policy or primary regulation.

Scientific review: Several mutually reinforcing papers may crowd out one contradictory result.

Enterprise search: Frequently accessed documents may dominate newer but less connected updates.

Support systems: Short troubleshooting passages may outrank a longer document containing the actual exception.

Conclusion

Reranking makes large retrieval systems usable by deciding what deserves scarce context. That decision is unavoidable and consequential.

The system remains answerable to reality only when ranking is treated as a contestable selection mechanism rather than an invisible proof of importance. What falls below the cutoff may still be what matters most.

Related Phrases and Concepts

- reranking
- cross-encoder reranking
- top-k retrieval
- context selection
- reciprocal rank fusion
- evidence bottleneck
- context budget
- representational authority

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)